Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
Instructor: Professor Sourin Das

A Term Paper Review of Chapters 1–3[1]

Based on the MIT Doctoral Thesis by Si Hui Tan (2010)

Authors

Sapnodip Pramanik
23MS145

Bikash Jyoti Saharia
23MS133

Bhargab Neog
23MS177

Shounak Goswami
23MS158

April 25, 2026

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, KOLKATA

Acknowledgements

[ACKNOWLEDGEMENT]

Abstract

[ABSTRACT]

Contents

1 Introduction **2**

2 Continuous Variable Quantum Information Theory **2**

3 State Discrimination Using Gaussian States **2**

3.1 Distinguisability Of States 2

3.1.1 Helstrom's Optimal Discrimination 2

References **4**

1 Introduction

2 Continuous Variable Quantum Information Theory

3 State Discrimination Using Gaussian States

3.1 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.1.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively. The error probability can simply be obtained from the hypothesis testing for H_0 and H_1 as

$$P_E = \frac{1}{2}(P(H_0|H_1) + P(H_1|H_0)) \tag{1}$$

where,
$P(H_0|H_1)$ is the probability of concluding that H_0 is true, when H_1 is and
$P(H_1|H_0)$ is the probability of concluding that H_1 is true, when H_0 is true and
the factor of $\frac{1}{2}$ arises from the equiprobable assumptions.

Also, note that the expectation of an operator $\hat{O}$ given the system is in state $\hat{\rho}$ is given by $\left\langle \hat{O} \right\rangle = tr(r\hat{h}o\hat{O})$. So, if the system is in $\hat{\rho}_j$ and we make a measurement with the operator with $\hat{E}_i$, then the probability of error in the measurement is given by

$$P(H_i|H_j) = tr(\hat{\rho}_j \hat{E}_i) \tag{2}$$

Thus, using eq 2 in eq 1, we get

$$P_E = \frac{1}{2}(tr(\hat{\rho}_1\hat{E}_0) + tr(\hat{\rho}_0\hat{E}_1)) \tag{3}$$

Also, since we defined our POVM to be the set $\hat{E}_0, \hat{E}_1$ and POVM are complete, so $\hat{E}_0 + \hat{E}_1 = I \implies \hat{E}_0 = I - \hat{E}_1$. So, eq 3 will be,

$$\begin{aligned} P_E &= \frac{1}{2}(tr(\hat{\rho}_1(I - \hat{E}_1)) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(tr(\hat{\rho}_1) - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(1 - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)), \text{since } tr(\hat{\rho}_i) = 1 \\ &= \frac{1}{2}(1 - tr((\hat{\rho}_1 - \hat{\rho}_0)\hat{E}_1)) \end{aligned} \tag{4}$$

Note, all we aim to is find the correct POVM set so that the error probability is minimum. For this case, as $\hat{E}_0 + \hat{E}_1 = I$, so, we can express $\hat{E}_0$ in terms of $\hat{E}_1$. Thus, we have to minimize P_E over the measurement operator $\hat{E}_1$. Define $\hat{\gamma} = \hat{\rho}_1 - \hat{\rho}_0$. Clearly, $tr(\hat{\gamma}) = tr(\hat{\rho}_1) - tr(\hat{\rho}_0) = 1 - 1 = 0$. Since, both $\hat{\rho}_0$ and $\hat{\rho}_1$ are both positive and semi definite, so the eigenvalues of $\hat{\gamma}$ can be both positive or negative. Since both $\hat{\rho}_0$ and $\hat{\rho}_1$ are hermitian, so $\hat{\gamma}$ is also hermitian, hence it has the spectral decomposition,

$$\hat{\gamma} = \sum_k \mu_k |k\rangle \langle k| \tag{5}$$

where μ_k are the real eigenvalues. Now expanding $\hat{E}_1$ in the same basis

$$\hat{E}_1 = \sum_{k,l} e_{kl} |k\rangle \langle l| \tag{6}$$

Hence,

$$\begin{aligned} tr(\hat{E}_1\hat{\gamma}) &= tr\left(\sum_{k,l} e_{kl} |k\rangle \langle l| \cdot \sum_m \mu_m |m\rangle \langle m|\right) \\ &= \sum_{klm} e_{kl}\mu_m \langle l|m\rangle \langle m|k\rangle \\ &= \sum_{klm} e_{kl}\mu_m \delta_{l,m}\delta_{m,k} \\ &= \sum_k e_{kk}\mu_k \end{aligned} \tag{7}$$

Since POVMs are positive valued, so e_{kk} is positive. Now, μ_k can be positive or negative, but since we are optimizing e_{kk}, if μ_k is negative, it will lead to negative propagation

of the errors, thus for such cases e_{kk} must be 0. Hence the net contribution will come only from the positive e_{kk} values. Thus the optimal $\hat{E}_1$ is one that projects onto positive part of $\hat{\gamma}$, $\hat{\gamma}_+$ (say) which projects to the subspace by the eigenstates of $\hat{\gamma}$ with positive eigenvalues.

References

[1] Si Hui Tan. "Quantum State Discrimination with Bosonic Channels and Gaussian States". In: *MIT Doctoral Thesis* (2010), pp. 0–70.